

Gradination: Hypothesis for Positioning and Patterning*

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SYNOPSIS. The interaction of contiguous proteins is explored in microtubules, rosettes, and membranes based on the well established molecular phenomena of cooperativity and allosterism. It is proposed that conformational gradients in protein arrays cause the formation of gradions by nearest-neighbor interactions. Gradions are repeating functional molecular sequences that contain several conformational forms of one or more proteins, with the result that different reactive sites can exist in the same molecular architecture at any one time. Gradionators are small controlling molecules that may be microscopically visible as layers or linkages, but could also be smaller. Some of the presently available supporting evidence and its functional implications are discussed, including the possibility that the *raison d'être* for membrane-particle arrays is to enhance the regulation and amplification capabilities of cell systems.

Index Key Words: Gradination; patterning; rosettes; membranes; microtubules; allosterism; cooperativity.

OUR discussion begins with an introduction that contains 2 parts: first a dedication and second a confession.

I have long been a devoted admirer of the protozoa though not as a protozoologist broadly acquainted with hundreds of organisms. Rather, I am a cell biologist and electron microscopist who has found protozoan cells to be particularly useful organisms for study. Thus I dedicate my talk to the protozoa, especially the heliozoa; and to protozoologists, especially Tracy Sonneborn, who taught and encouraged me much over the last 28 years and whose "retirement" this year heralds, I hope, a new burst of research activity.

Now for the confession. I would like to say that we have now firmly established all that is presented here. We are, however, more in the position of which John Bunyan wrote in the early pages of *Pilgrim's Progress* when he described the initial plight of Pilgrim who was asked whether he saw the wicket gate, the ultimate goal that he sought. You will remember his inability to see it and his confession that he could not even see "yonder shining light" with certainty. He was nevertheless admonished to utilize what light he had and to proceed on his quest. So I cannot claim to take you to the gate; I can only invite you to hear about the stages of a pilgrimage that attempts to understand the patterns in a large field.

Our presentation is divided into 5 parts. The first 2 often seem cramped, small, unrelated areas on the path. The 3rd begins to show some light, the 4th is a long tunnel, and the 5th is a huge field with contours poorly defined and variously arranged. So let us start down the pilgrim's path.

I. *Monomeric Enzymes.*—Simple enzymes usually have molecular weights of 20,000 to 90,000 and a single active site that combines with only 1 substrate molecule. These active sites are shaped in accordance with the form of the molecule to be affected and are usually described as crevices, shallow depressions, or pits; pits tend to be for the cleavage of a single end amino acid from a protein, whereas crevices and depressions tend to be found where long molecular strands are to be cut. In these sites, several specific amino acids are the critical points for globular proteins such as the carboxypeptidases, chymotrypsin, lysozyme, the ribonucleases, and papain.

These enzymes may not be so simple in some cases. They may require metal ions, other organic molecules, or rather specific cofactors to operate. Frequently, reduced enzymic action results from the occupancy of the active site by molecules that are refractory to enzymic action, a phenomenon that is called "competitive inhibition."

Plainly these monomeric enzymes, that is enzymes composed of 1 polypeptide chain, are functionally regulated or threatened if mutations alter their amino acid components, if required metal ions or cofactors are lacking or limiting, or if competitive inhibitors are active from either *in vivo* or foreign sources. Their interest to us in considerations of patterning lies in helping us understand protein functions as currently known; though they are enzymic proteins, their properties are in large part indicative of the ways that all proteins are regulated, if it is understood that the active sites are binding sites but not always cleavage sites for the interacting molecular species.

Specific illustrations can be found in any biochemistry text. Diagrammatically, they are illustrated in Fig. 1.

II. *Oligomeric Enzymes.*—More complex phenomena result when enzymes are composed of several subunits. The resulting properties include those described for simple enzymes, but added properties also arise from the way the components are organized and then interact. Many—perhaps most—enzymes are complex in this way, have molecular weights usually ranging from 35,000 to several million, and may be composed either of homo- or heteropolymers. Coenzymes are present, and only after their binding can the substrate be bound; interestingly, 1 coenzyme molecule can now "turn on" the binding capabilities of more than 1 subunit in the molecule. Thus a small coenzyme molecule in comparatively short supply controls the function of a much larger population of larger molecules.

Some of the better known examples of such enzymes are the dehydrogenases, and one of the best studied is glutamate dehydrogenase (13). It is now thought to be composed of 6 identical subunits, each of ~ 55,000 molecular weight; the molecule thus has a molecular weight of over 300,000 daltons, a length of 100-130 nm, and a diameter of ~ 80 nm. *In vitro* methods can induce chains, sheets, and tubes that appear in high-resolution electronmicrographs to be composed of numerous X-shaped polymerized molecules. Two types of binding sites are known; one set is blocked by polymerization, whereas the other is not.

Another, somewhat more complex, enzyme is aspartate transcarbamylase. It is a regulatory enzyme that not only catalyzes a specific reaction but also responds to the cellular levels of relatively unrelated metabolites. It has been shown recently (23) that this enzyme is composed of 12 subunits: 6 regulator

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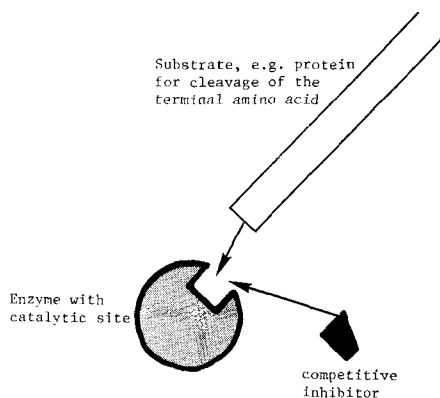


Fig. 1. Schematic diagram of the interaction of a substrate with a simple monomeric enzyme. Note that the active site, which is highly structured and therefore quite specific in its activity, can be occupied competitively by a substrate or by one or more alternative inhibitor molecules.

units, $\sim 34,000$ daltons each, are present as 3 dimers, whereas the 6 catalytic subunits are $\sim 100,000$ daltons each, occupy central locations, and are in the form of 2 trimers. In addition, 6 zinc atoms are present in the regulatory subunits.

Best studied of all proteins of this type is hemoglobin which has subunits, 2 α and 2 β amino acid chains (17). And so the story goes on to many, many more oligomeric enzyme patterns that have been elucidated in the past 15 years.

Now why are enzyme subunit arrangements so useful for us to understand here? For our considerations of patterning, there are 2 compelling reasons that are based on 2 attributes of these systems.

First, when subunits are aggregated in these precise forms, they are known to cooperate. "Cooperativity" is a term which simply says that if one subunit "does its thing," another subunit in the molecule can "do its thing" *much more easily*. For example, in hemoglobin, if conditions exist for one O_2 molecule to be bound to one subunit, this binding causes the remaining subunits to be changed so that they more easily bind O_2 molecules. This example illustrates positive cooperativity; negative cooperativity has also been demonstrated in oligomeric enzymes.

Second, many enzymes are regulated by effectors that are bound at sites other than the active catalytic site. These effects are called allosteric mechanisms, and the key to them is a change in subunit shapes or conformations. Simply stated, a subunit has 2 binding sites; one is the active site for its enzyme action and is controlled—turned off, on, or partially on—by what is happening at the other site. In fact, effector molecules can occupy an allosteric site and change not only the catalytic-site activity on that subunit but also on adjacent subunits. The regulatory nature of such a system is really impressive!

Add to this system the possibility of competition by more than one molecule for both the catalytic and allosteric sites—a phenomenon known as competitive inhibition—and the range of normal control processes is visualized. This system is diagrammatically illustrated in Fig. 2. You may also want to try to visualize the places where pathological alterations are possible by mutant molecules or foreign substances.

So we end our consideration of parts I and II. The information included in these parts is quite well established on the basis of elegant studies by many different groups whose work

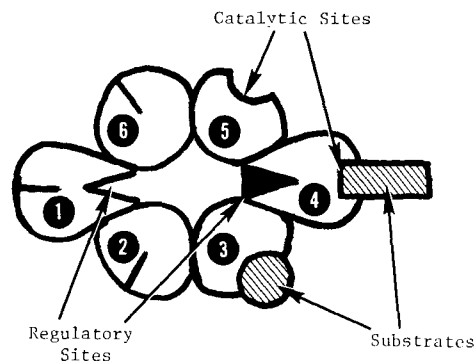


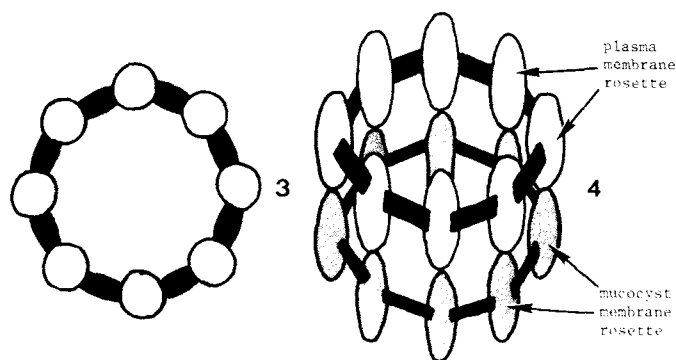
Fig. 2. Schematic diagram of an oligomeric enzyme composed of 2 elongated subunits and 4 more spherical subunits. Subunit 1 has an open and unoccupied regulatory site and, as a result, a closed catalytic site; subunit 4 has an occupied regulatory site with the result that its catalytic site was opened and has been occupied; this example indicates the changes that result by allosteric intramolecular action. *Intermolecular* action is illustrated in subunits 2, 3, 5, and 6; the binding of a molecule at the catalytic site of subunit 4 is said to result in a change of the conformation of subunit 5 which now has its catalytic site made active (in contrast to subunits 2 and 6) and of subunit 3 which now has both an active and occupied catalytic site. "Cooperativity" is the term applied to this intermolecular allostereism.

can be touched upon only lightly today (see Refs. 8 or 16 for further information). With the concepts "cooperativity" and "allosterism" plainly before us, let us begin to look at some cell patterns.

III. *Rosettes*.—Rosettes are ordered rings of membrane-embedded particles reported during the past several years from a few protozoan species examined by the freeze-fracture method. These ring patterns are composed of 7-12 granules that are essential features in areas of membrane fusion; that is, where, according to Bardele (2), kinetocysts are extruded in the centrohelidan heliozoa and where host and prey membranes fuse in suctorian feeding. Bardele suggested that these rosette particles were connected by membrane proteins. If so, the situation could be diagrammed as shown in Fig. 3. Now, if we utilize the concepts of cooperativity and allosterism, we can see that triggering of this complex need not require contact with all particles. Stimuli at only 1 or a few subunits followed by allosteric changes in the rest could cause total activation of the complex.

The most detailed structural study of rosettes is that of Satir et al. (22). They show that rosettes are found in both plasma membranes and mucocyst membranes of *Tetrahymena*. In fact, the 2 rosette systems are in register with each other (Fig. 4). Stimuli conceived as the induction of a conformational change by binding of an effector molecule at an external site cause allosteric, cooperative changes in the conformations and functional states of all 16 functional subunits. It may be particularly noteworthy that studies of mutant organisms suggest that 5 particles are required before rosettes are functional; it can be inferred that cooperativity between particles is obligatory in this system (4). Although we do not know all of the structural features of rosettes, we do have possible ways of thinking in detail about their method of functioning by applying the concepts of cooperativity and allosterism.

To summarize to this point, we can begin to see some ad-



Figs. 3, 4. [Interpretive diagrams of rosettes.] 3. Membranes have been shown to include particles 12 μm in diameter uniformly spaced in a circular shape. Patterning is postulated to be based on linkages that are intermediating molecules through which cooperativity would operate. 4. Three-dimensional form of rosettes which are typically found where 2 membranes are apposed intimately and fuse when functionally activated. It is conceived that an external signal induces an effect that activates the assembly.

vancing complexity and an application to a simple system. We can also see some distances of actions of these systems (Table 1).

IV. *Microtubules*.—Those of you who know my research realize that microtubules have fascinated me from the time I saw them in hypotrichs and flagellates during my Ph.D. dissertation research and through the 20 years since. Many biologists have now added much to our understanding of microtubules, and their work is summarized in a recent New York Academy meeting, the proceedings of which include over 800 pages (24). For our purpose the salient features of current microtubule knowledge are the following.

First, microtubules are composed of 2 subunits, termed α and β tubulins, which form dimers and then polymers that are essentially infinitely long—perhaps a million dimers. They can be thought of as protofilaments that are end-to-end polymers of dimers which are then bound together to form a tube with an open lumen. The cross sections of axopodia in the giant heliozoon, *Echinospaerium nucleofilum*, are good examples (Fig. 5). Studies of microtubule cross sections in our laboratory using

standard rotation and superpositioning methods have shown that both 12 and 13 subunits can be demonstrated (Fig. 6). Our studies of over 75 cross sections have revealed few cases of 11 subunits and dozens of cases with either 13 or 12 protofilaments. Studies by Tilney and others using a different method in which the cell is abused enough to allow tannic acid molecules to enter the cell freely and to coat the microtubules in a negative-stain method, have shown 13 protofilaments (26). In yet another method natural patterns are used to infer protofilament numbers. Curving or zig-zag rows would be found in some organisms (6, 12) and would usually indicate odd protofilament numbers; even protofilament numbers would be inferred from straight rows, as seen at places in Fig. 5. Whether the proper number is 12 or 13 or whether microtubules can form with varying numbers of protofilaments as suggested by Burton et al. (5), is not certain; I am quite prepared to believe that microtubules can have their architecture reversibly altered for various functional purposes, but data to that effect are not available. Nevertheless a microtubule can plainly be understood as a highly oriented pattern of tubulin dimers.

Second, the dimer has now been studied biochemically enough that a recent report suggests 3 different conformations; the work by Ludueña and coworkers (15) based on studies of binding to isolated dimers, has allowed him to diagram a proposed dimer structure (Fig. 7 and personal communication). Colchicine is seen to bind to one common dimer site; this example of competitive inhibition no longer allows the dimer to polymerize into microtubules. Other binding sites are paired, 1/tubulin molecule.

Third, microtubules are known to have variable stability. For example in mitosis, microtubules that can be degraded by cold treatment during most mitotic stages are coated by a dark substance, included in the midbody at telophase, and are then refractory to cold treatment (11, 21). The implication is that surface coatings can cause stability changes in polymerized microtubules.

Fourth, although microtubules are thought to grow by the addition of individual dimers at the ends of microtubule segments (10), degradation can proceed at rates so fast that "cataclysmic" is an appropriate descriptive word. It is thus necessary to think that microtubules, although relatively stable in their functional forms, are in a condition that allows attack at many places along their lengths (20).

TABLE 1. *Progression of patterning and function in cellular systems.*

Examples	Regulatory type	Proposed regulatory capabilities	Estimated distance of action (μm)
Monomeric enzymes	Simple model	Small-molecule control of enzymatic activity.	0
Oligomeric enzymes	Confined, closed fields	Range of conformational states in one functional unit. Allosteric small-molecule control of large populations of large molecules. Amplification of small-molecule control by cooperativity.	0.02
Rosettes	Confined, open fields	All of the above. External stimuli cause membrane changes and signal internal functions.	0.1
Microtubules	Extensive, fully packed, gradionated fields	All of the above. Repetitious field production. Control of linear organization. Position determination. Provision for a range of stability to stimuli.	0.5
Membranes	All types of gradionated fields above and also extensive, open gradionated fields	All of the above. Organizations of membrane coatings. Transmembrane communication.	at least 3

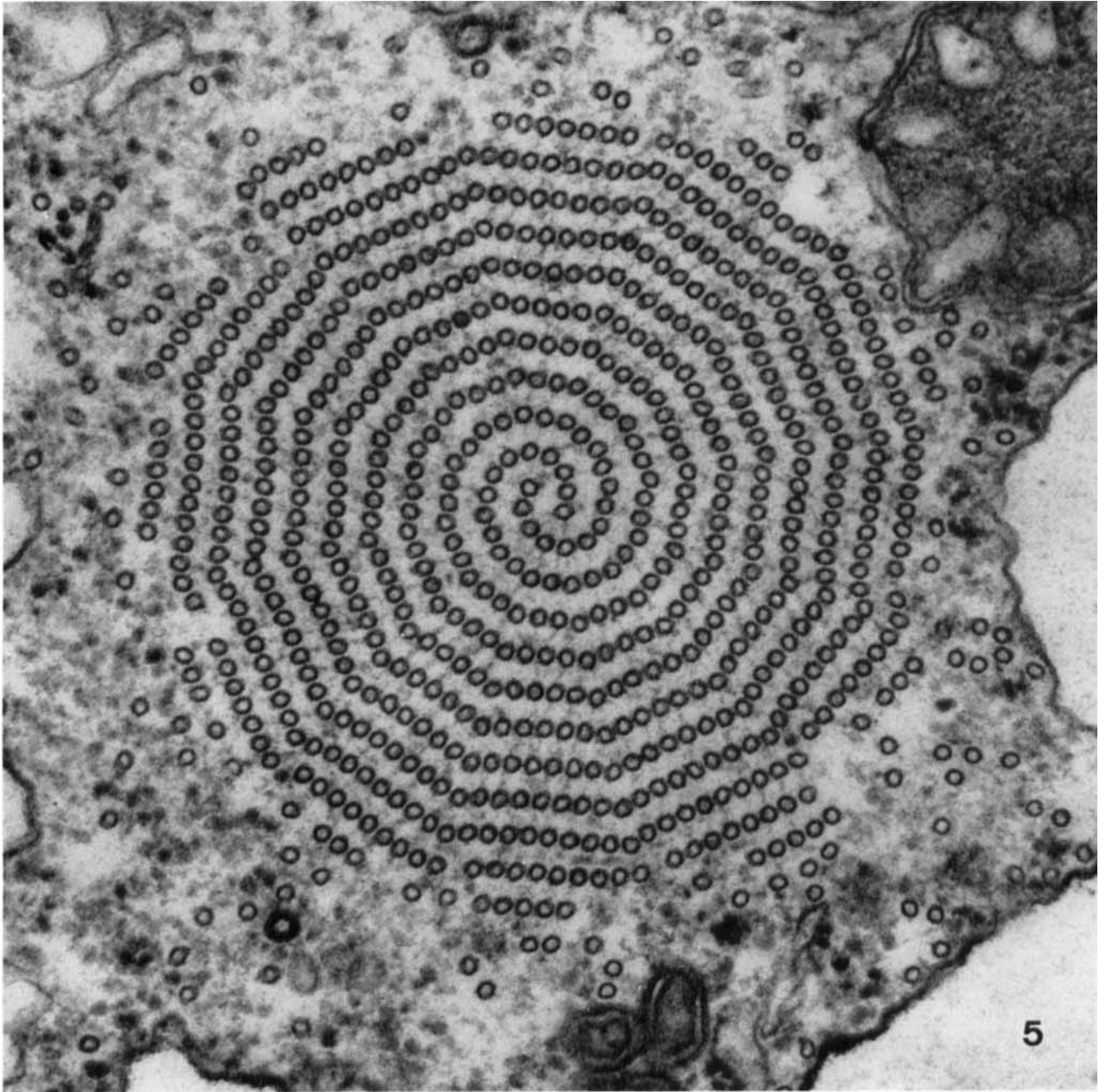


Fig. 5. Cross section of the axoneme of the giant heliozoan, *E. nucleofilum*. The microtubule pattern consists of numerous straight rows toward the periphery of the 2 spiraling "sheets" of microtubules. $\times 110,000$.

Fifth, the controlled patterns of microtubule aggregates are striking in their regularity. The heliozoa are the best examples of such precision, and during this symposium Christian Bardele will elaborate on that striking order and several beautiful patterns. The Cachons (6) have discussed these patterns in detail. Linkage structures are present in such arrays, and it can be inferred that microtubule cross-bridges are instruments of pattern determination. In *Echinospaerium*, the pattern which is so precise can be explained on the basis of 2 spacings attributed to 2 linkages (Fig. 8)—one short and one long—that can be modeled into providing a complete explanation of the array as seen in cross-section (Fig. 5).

It is not enough, however, to speak only of 2 linkers; it is necessary to consider a mechanism that specifies the exact placement of those linkers both spatially and temporally to give the demonstrated pattern and only that pattern. If a summary of occurrence of short spacings when 2 short linkages are already present is made, a frequency of linkage is defined (Fig. 9). To achieve the desired pattern, a very strong preference must be given to bonding exactly opposite from the first short linkers; furthermore, a graded frequency of bonding down to zero possibility is necessary since microtubule patterns are never seen in this organism where adjacent pairs of short linkages would be required. Thus a circumferential gradation of bonding fre-

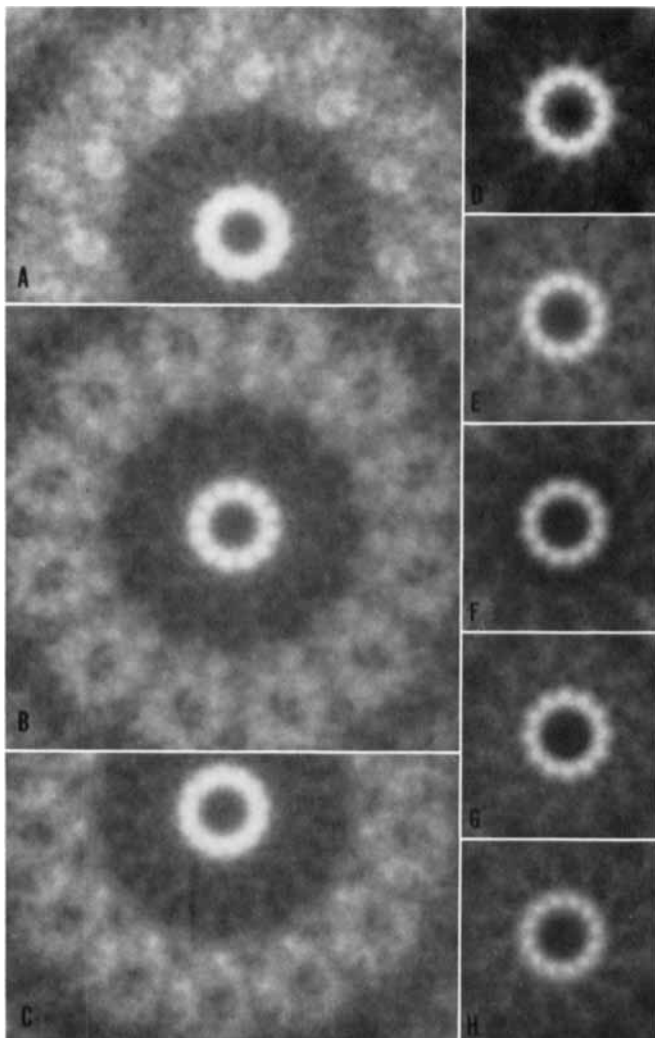


Fig. 6. Markham rotations of microtubules from *E. nucleofilum*. A-C are rotations of one of the centermost microtubules in the axoneme; A is rotated for 11-protofilament symmetry, B for 12, and C for 13. Note that not only is the 12-fold symmetry clearly shown in B, but that each protofilament aligns well to the microtubules in the first row. D-H are additional examples of 12 protofilaments.

quency is shown. Similarly, along the lengths of these microtubules, as in many other microtubules, linkages are found only at intervals that reflect bonding at approximately every 4th dimer. The restraints on this system are extraordinary and call for repetitious and regular spacing of elements at precise dimer intervals.

Such requirements caused us to propose in 1970 that recurrent gradients exist in microtubules and are determined by the bonding of linkage elements (19). The best way to explain this phenomenon is to use the concepts of allosterism and cooperativity that have been established for enzymes. Such a unit of repetitious structure was termed a gradion and, using the dimer model of Ludueña, can be diagrammed in one protofilament (Fig. 10).

Five conformational states are assumed to exist. Number 1 designates the suggested conformation resulting from dimer polymerization; later, once an intermicrotubule linkage has been made, that subunit changes to a different conformation, indi-

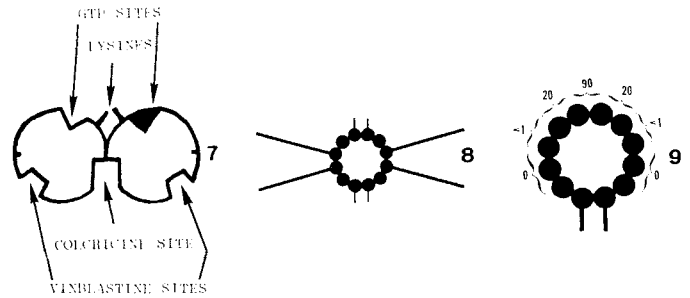


Fig. 7. Schematic diagram of the tubulin dimer modified from Ludueña (personal communication). Note that in contrast to vinblastine- and GTP-binding sites said to be present on each subunit, the colchicine site is thought to be shared by both.

Fig. 8. Diagram of a cross section of an axonemal microtubule from *E. nucleofilum*. The maximum number of linkages indicated allows an orderly model of the entire axoneme cross section to be made.

Fig. 9. Probability of linkage in the *E. nucleofilum* axoneme. If a pair of short linkages has been made, the probability ratios for additional short linkages are indicated and have been derived by actual tabulations of angles between microtubules. Note that by far the highest probability is for binding exactly opposite to the original pair. No cases of binding at subunits adjacent to the original pair have been noted.

cated by number 5, and has a nearest-neighbor effect to alter the adjacent dimer conformation which sequentially alters its neighbors to form a gradient. The space between 2 such linkages then becomes a gradion. A zero designation is given for a dimer not yet polymerized, because it is apparent that bonding of linkers must be completely excluded in the unpolymerized state.

The dimer model proposed by Ludueña gives us a basis for considering these phenomena further. Of the sites available, the colchicine site or some other single dimer-affecting site is the most attractive since it would control the whole dimer most efficiently. Therefore let us refer to a "gradination site" and consider that it is allosterically controlled by binding at other sites of the dimer and that it can be competitively occupied by a number of different native substrates or foreign molecules (Fig. 11).

It must be emphasized that a microtubule cannot be functionally understood on the basis of dealing with a protofilament. The more complex arrangement of a full microtubule can be visualized as though it were an opened and flattened field of dimers; then we can diagrammatically work with it in a 2-dimensional format such as we had to use for illustrations here. Let us also suppose that each dimer is simply represented by a numbered circle and that both types of linkage in the heliozoan axoneme are designated by sixes and fours respectively. If gradions are drawn in all directions from short-linkage dimers, based on a descending nearest-neighbor effect, we have gradination fields (Fig. 12). Thus if linkages are completed at positions numbered 6, several adjacent dimers are "told" that they may no longer accept either long or short linkages. Thus, a mechanism for the proper placement of given dimers can be developed by suggesting that linkages of one type, probably the short ones, are preferred in conformation 1, and the other is preferred in conformation 4.

The result is that, even though high numbers are induced over much of the microtubule, there are still areas where low numbers appear infrequently and can be considered as places of attack for cataclysmic degradation despite full gradination. Competition for binding at all gradination sites can also be considered on both positive or negative-inhibition bases. The formulation

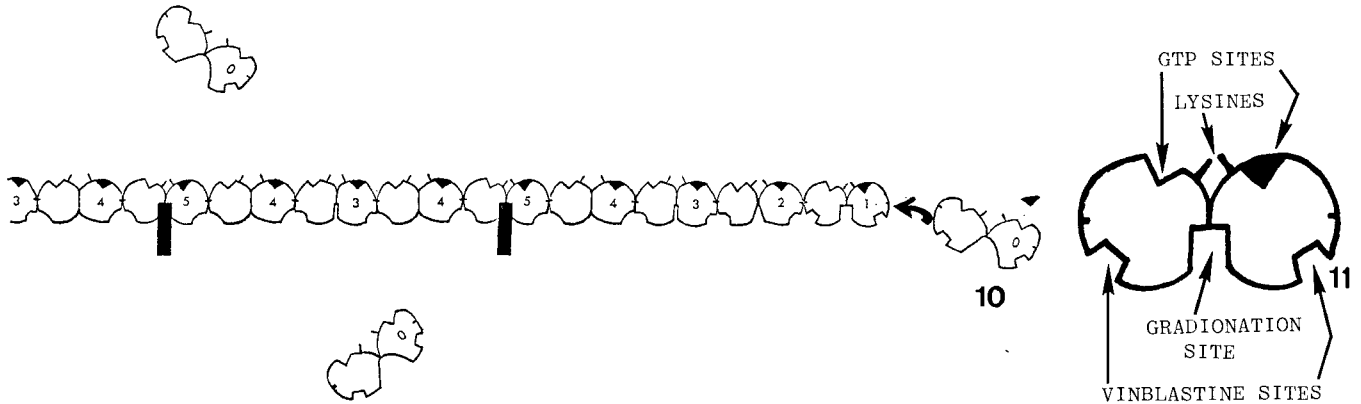


Fig. 10. A gradinated protofilament based on 5 conformational states of polymerized tubulin dimers. State 5 is suggested to be induced by a linkage with a gradinator molecule; such a state has a graded, nearest-neighbor effect to preclude linkages at adjacent dimers. One gradion would include 2 dimers that contain gradinator-linkers and 3 interposed dimers that include 2 different conformations. Another conformational state, indicated by "0," seems to be necessary and would preclude gradinator bonding but allow an initial, loose colchicine bonding that could later become tight and induce a different conformation in free dimers.

Fig. 11. This dimer diagram is based on the premise that a single site, here hypothesized to be similar to the colchicine site (Fig. 7), can be competitively occupied by a variety of gradinator molecules, only some of which would crosslink between microtubules. Other such molecules could be very small and submicroscopic but still able to change dimer conformations. This site is termed the "gradination site."

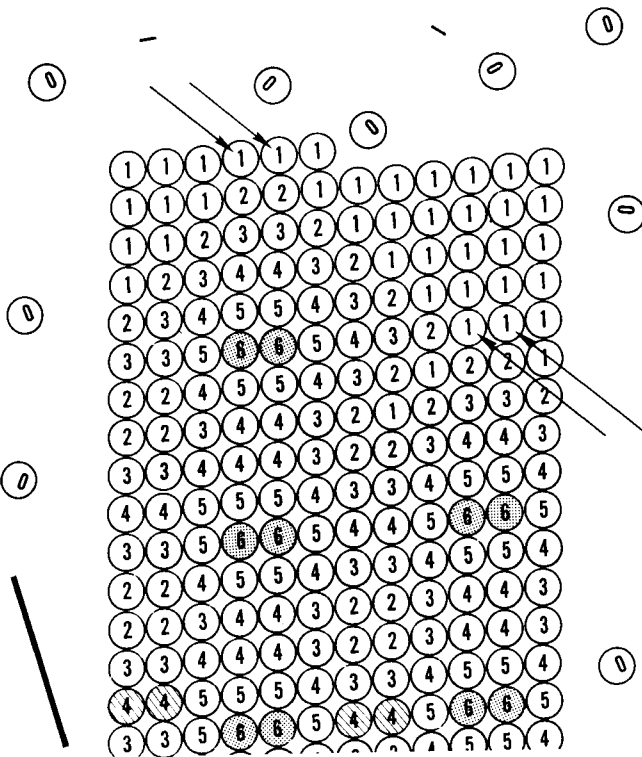


Fig. 12. Schematic diagram of a flattened, opened microtubule. Gradination fields would be formed whenever a short-linkage pair binds to 2 favored dimers (arrows), each of which is changed to a conformation indicated by a circle numbered 6; binding would result in nearest-neighbor effects in all directions and would not only preclude binding at adjacent dimers but leave some conformations such as 1 and 2 at some locations, even when a 6-conformation gradination is postulated. Long-link locations would be favored at positions numbered 4. Geometric constraints would make linkages at both "6" and "4" sites either impossible or inoperable if on the wrong protofilaments. Varying numbers of protofilaments and pitches of subunit bonding could easily be adapted to the phenomenon postulated. Long and short linkage components of the system are drawn to approximate scale as solid bars.

therefore assumes both a graded cooperativity and a graded allostery applied to essentially an infinite, fully packed array of subunits.

The result of this process would be the formation of many, many repetitious fields, each of which includes a variability of stable dimers, a mechanism for position determination, and a mechanism for forming a precise array. Furthermore, control of a high number of comparatively large dimers can be envisioned by a comparatively small population of various determinant smaller molecules—gradinator molecules—which can be produced rather quickly under genetic control.

Considerations such as this with regard to microtubules allow us to offer hypotheses for mechanisms that give rise to positioning and movement phenomena such as parallel and perpendicular arrays of microtubules, the procentriole spacing in both orientation and distance from a parent centriole, the sliding of microtubules past each other, and the movement of vesicles attached by linkage structures to microtubules. Gradination in this sense are expected to be visible as linkage structures, arms, or layers of material in electronmicrographs; it is possible however that gradination are present which are not thus visible but still function *in vivo*. Obviously not only microtubules but neurotubules, microfilaments, rosettes, and other fibrous organelles can be similarly visualized.

Thus we are able to begin to deal with a few pattern and position phenomena seen in cells. Let us now proceed with the concepts of cooperativity, allostery, and gradination as applied to considerations of membranes.

V. *Membranes*.—In the introduction, I referred to the 5th part of this talk as a huge field with contours poorly defined and variously arrayed. It might be best in fact to describe it as a "slough of despond" with many pilgrims mired in difficulty.

Much attention has been given to understanding membranes at the level of lipid bilayers and to proteins included therein, but our understanding of inner and outer surface coatings is based largely on gross collections of molecules analyzed for major components by biochemical procedures. There is reason to believe that a membrane surface could be visualized in a fashion similar to that diagrammed here for an opened microtubule (Fig. 12) with the exception that the area should be extended

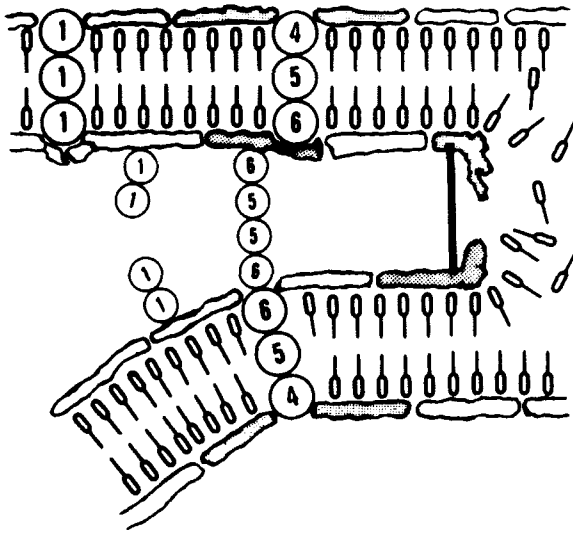


Fig. 13. Trans- and inter-membrane gradions are diagrammed in ways similar to those needed to explain the functions of rosettes. A series of subunits with conformation 1 are changed to conformations 4, 5, and 6 when bonding occurs between surface proteins and cooperativity causes changes in adjacent subunit conformations. Not only is it suggested that spherical subunits are changed, but elongated surface molecules are also modified (stippled rods). Linkages (solid bar) could also bind to surface molecules changing their conformations and active sites.

to an essentially infinite protein array capable of similar gradination. Cooperative and allosteric phenomena operating along membrane surfaces could then be conceived; furthermore, the same mechanisms could function into and through membranes. It may be unnecessary to think of fully packed fields in such systems, but merely to suggest that networks could be composed of elongated molecules; thus considerably greater distances could be gradinated and functionally affected along membrane surfaces.

The best model system to depict membrane structure-function relationships, however, is the rosette. Here an ordered membrane domain can be visualized and correlated with the control of membrane properties for on-demand membrane fusion in several protozoan cell functions (2, 22). Communication and patterning across, between, and along surfaces of both membranes have been demonstrated. Similarly, the "patch," seen in protozoan ciliary membranes, that may be correlated with calcium physiology (3) is another intramembrane particle array. These models epitomize simple patterns where cell-surface architecture is correlated with intracellular structures and functions. Amplifications of such models by the addition of multiple radiating gradions to specify large patterns and structural elaborations are rather simple to visualize (Fig. 13; this diagram was made in 1969, several years before rosettes were known. Although the drawing could be improved now, it is essentially a prediction of the rosette functions now known.)

Large fields of regularly spaced particles are being seen almost ubiquitously as freeze-fracture techniques are being widely used. Guinea-pig sperm have been shown to have numerous regions of such particles, and the suggestion has been made that they are involved in the correlation of organelle function from one developmental state to another (14). Particle aggregates and "paired doublet arrays" have been observed in excitable membranes (9). Cell junctions have been shown to have a network of particle-rows in a variety of cells; in particular, studies

of both striated muscle (e.g. 18) and smooth muscle (e.g. 7) have been carried to the extent that particle spacings can be cited for several species (18). Numerous studies on other tissues from several laboratories could be cited, and the reader is referred to almost any of the recent issues of cell-structure oriented journals for more examples. The reader should, however, not miss the obvious possibility that allosterism and cooperativity in these systems would greatly facilitate their functioning. In fact, allosterism and cooperativity may well prove to be the *raison d'être* for the existence of such particles and their arrays in membranes. The regulation of numerous cell systems could be enhanced and "fine-tuned," and, in addition, the amplification of "signals" could be achieved, all in the context of trans-membrane communication. By using the mucocyst rosette as a model, the application to numerous systems can be hypothesized with a molecular basis in this context.

Although we must know much more about membranes, their included molecules, and their coatings, we perhaps begin to understand enough to consider patterning problems, many of which have dealt with the pellicles of protozoan cells. These complex surface elaborations contain not only membranes but also microtubules, microfilaments, surface coatings, layers of material called epiplasm, and included organelles such as cilia. In such a complex, it is easy to conceive that the pattern of the *Paramecium* pellicle and its anomalies, which have been so carefully studied and recently discussed by Sonneborn (25) and described in added detail by Allen (1), are determined by gradions and gradination fields; these, determining features, originating from a marker organelle such as a cilium and radiating out along many structures could prescribe, in successive stages, the placement of other structures and the limits of repeating units. It is our contention that we are just beginning to find ways to describe these mechanisms, even to the extent of dealing with large cell surfaces by conceiving of several levels and many replicates of gradination fields.

A summary of the 5 areas discussed is presented in Table 1. Notice the increasing span of distances that can be conceived as being covered as the systems are elaborated. We propose that patterning is affected by small numbers of determinant molecules allosterically regulating contiguously placed—perhaps polymerized—molecules that occur in comparatively high numbers. Gradination, which is dependent on several *in situ* or polymerized conformations and a distinctive conformation of unpolymerized molecular species, is offered to explain patterning phenomena and to maximize functional controls. Gradinationators are viewed as competitive inhibitors that can exist in great variety in normal processes and that may also be foreign substances causing pathological conditions.

I hope that you can now understand a little of the excitement—and risk—that we have felt as pilgrims on a path that leads to many areas and to a considerable universality of application. Certain predictors of cellular order are indicated; many functional mechanisms and their pathologies are also suggested. Time precludes my further discussion of such matters here, and I close by inviting you to join us on the pilgrim road.

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Organization and Control of Microtubule Pattern in Centroheliidan Heliozoa*†

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SYNOPSIS. Comparative studies of axopodial microtubule pattern in 10 different centroheliidan Heliozoa belonging to the genera *Acanthocystis*, *Raphidiophrys* and *Heterophrys* suggest that 2 basic principles govern pattern formation in centroheliidan Heliozoa. While the larger "open" arrays with unspecified number of microtubules, e.g. in *A. aculeata* and *R. ambigua*, may result from self-linkage of additional microtubules around centroplast-nucleated "starter microtubules," the smaller "closed" arrays with specified microtubule number, e.g. in *A. pectinata* and *H. marina*, favor a template-driven linkage mechanism. The centroplast is a highly complex microtubule organizing center involved in the control of orientation, number, and diameter of the axonemes. Its shell may serve as a surface upon which the microtubule nucleating sites assemble, but how the precise positioning of these sites occurs is still open to debate. Some of the unsolved problems of microtubule pattern formation may be explained by the "linker nucleation hypothesis" which is an extension of the "gradion hypothesis" by Roth et al. It is shown how both the formation of closed arrays and the balanced lateral growth of open arrays may result from linker-induced microtubule nucleation.

Index Key Words: Heliozoa; Centroheliida; axopodia; microtubule patterning; microtubule organizing center; linker-induced microtubule nucleation.

PATTERN formation in protozoa comprises events of widely differing complexity which range from self-assembly of a restricted number of different macromolecular subunits to form

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comparatively simple structures such as a rhizoplast, a pelta, or a trichocyst to the highly complex spatial arrangement of a large variety of different organelles and structures in the polymastigine or ciliate cortex. The microtubule (mt) arrays found in heliozoan axopods have been shown to be a suitable model system for studying some of the supposedly simpler problems of pattern formation at the cellular level. The major question is whether the information for a particular mt pattern resides in the geometry of both the microtubules and the intertubular linkers, henceforth called pattern formation by *self-linkage*, or whether the microtubule organizing center (MTOC) (13) in